

3.2 Functions of Two or More Variables Worksheet

1. Let $f(x, y) = \frac{xy^2}{3x^2 + y^4}$. Determine the limits in the problem below. Make sure you write a full argument to justify your position.

- a) Determine $\lim_{y \rightarrow 0} f(y^2, y)$.

$$\text{We calculate } \lim_{y \rightarrow 0} f(y^2, y) = \lim_{y \rightarrow 0} \frac{y^2 y^2}{3(y^2)^2 + y^4} = \lim_{y \rightarrow 0} \frac{y^4}{4y^4} = \lim_{y \rightarrow 0} \frac{1}{4} = \frac{1}{4}.$$

- b) Determine $\lim_{(x,y) \rightarrow (0,0)} f(x, y)$.

If we progress towards $(0,0)$ along $y = x$ we see that

$$\lim_{(x,y) \rightarrow (0,0)} \frac{xy^2}{3x^2 + y^4} = \lim_{(x,x) \rightarrow (0,0)} \frac{x(x)^2}{3x^2 + (x)^4} = \lim_{x \rightarrow 0} \frac{x^3}{3x^2 + x^4} = \lim_{x \rightarrow 0} \frac{x}{3 + x^2} = \frac{0}{3} = 0.$$

However, if we progress towards $(0,0)$ along $x = y^2$ (as we did in part (a)) we see that

$$\lim_{(x,y) \rightarrow (0,0)} \frac{xy^2}{3x^2 + y^4} = \lim_{(y^2,y) \rightarrow (0,0)} \frac{xy^2}{3x^2 + y^4} = \lim_{y \rightarrow 0} \frac{y^2 y^2}{3(y^2)^2 + y^4} = \lim_{y \rightarrow 0} \frac{y^4}{4y^4} = \lim_{y \rightarrow 0} \frac{1}{4} = \frac{1}{4}$$

Since this limit has different answers as we progress towards $(0,0)$ along different paths, we can say that $\lim_{(x,y) \rightarrow (0,0)} f(x, y)$ DNE.

- c) Determine $\lim_{(x,y) \rightarrow (1,0)} f(x, y)$

We clearly see that at $(1,0)$ $f(x, y)$ is continuous and therefore we calculate

$$\lim_{(x,y) \rightarrow (1,0)} f(x, y) = \lim_{(x,y) \rightarrow (1,0)} \frac{xy^2}{3x^2 + y^4} = \frac{1(0)^2}{3(1)^2 + (0)^4} = \frac{0}{3} = 0$$

2. Determine whether each of the following limits exists. Make sure you write a full argument to justify your position.

a) $\lim_{(x,y) \rightarrow (0,0)} \frac{5x^2y^2}{x^4 + y^4}.$

If we progress towards $(0,0)$ along $y = x$ we see that

$$\lim_{(x,y) \rightarrow (0,0)} \frac{5x^2y^2}{x^4 + y^4} = \lim_{(y,y) \rightarrow (0,0)} \frac{5y^2y^2}{y^4 + y^4} = \lim_{y \rightarrow 0} \frac{5y^4}{2y^4} = \lim_{y \rightarrow 0} \frac{5}{2} = \frac{5}{2}.$$

However, if we progress towards $(0,0)$ along $y = 0$ we see that

$$\lim_{(x,y) \rightarrow (0,0)} \frac{5x^2y^2}{x^4 + y^4} = \lim_{(x,0) \rightarrow (0,0)} \frac{5x^2(0)^2}{x^4 + (0)^4} = \lim_{x \rightarrow 0} \frac{0}{x^4} = \lim_{x \rightarrow 0} 0 = 0$$

Since this limit has different answers as we progress towards $(0,0)$ along different paths, we can say

that $\lim_{(x,y) \rightarrow (0,0)} \frac{5x^2y^2}{x^4 + y^4} DNE.$

b) $\lim_{(x,y) \rightarrow (0,0)} \frac{\sin(\sqrt{x^2 + y^2})}{\sqrt{x^2 + y^2}}$

This limit is difficult to evaluate at first. So, if we let $r = \sqrt{x^2 + y^2}$ then as $(x,y) \rightarrow (0,0)$ clearly

$$r \rightarrow 0. \text{ Therefore, } \lim_{(x,y) \rightarrow (0,0)} \frac{\sin(\sqrt{x^2 + y^2})}{\sqrt{x^2 + y^2}} = \lim_{r \rightarrow 0} \frac{\sin(r)}{r} \stackrel{H}{=} \lim_{r \rightarrow 0} \frac{\cos(r)}{1} = \cos(0) = 1$$

c) $\lim_{(x,y) \rightarrow (0,0)} \frac{y^4 \cos^2(x)}{x^4 + y^4}$

If we progress towards $(0,0)$ along $y = 0$ we see that

$$\lim_{(x,y) \rightarrow (0,0)} \frac{y^4 \cos^2(x)}{x^4 + y^4} = \lim_{(x,0) \rightarrow (0,0)} \frac{(0)^4 \cos^2(x)}{x^4 + (0)^4} = \lim_{x \rightarrow 0} \frac{0}{x^4} = \lim_{x \rightarrow 0} 0 = 0.$$

However, if we progress towards $(0,0)$ along $x = 0$ we see that

$$\lim_{(x,y) \rightarrow (0,0)} \frac{y^4 \cos^2(x)}{x^4 + y^4} = \lim_{(0,y) \rightarrow (0,0)} \frac{y^4 \cos^2(0)}{(0)^4 + y^4} = \lim_{y \rightarrow 0} \frac{y^4}{y^4} = \lim_{y \rightarrow 0} 1 = 1$$

Since this limit has different answers as we progress towards $(0,0)$ along different paths, we can say

that $\lim_{(x,y) \rightarrow (0,0)} \frac{y^4 \cos^2(x)}{x^4 + y^4} DNE.$

d) $\lim_{(x,y) \rightarrow (0,0)} \frac{xy}{(x^2 + y^2)^2}$

If we progress towards $(0,0)$ along $y = 0$ we see that

$$\lim_{(x,y) \rightarrow (0,0)} \frac{xy}{(x^2 + y^2)^2} = \lim_{(x,0) \rightarrow (0,0)} \frac{x(0)}{(x^2 + (0)^2)^2} = \lim_{(x,0) \rightarrow (0,0)} \frac{0}{x^4} = \lim_{x \rightarrow 0} 0 = 0.$$

However, if we progress towards $(0,0)$ along $y = x$ we see that

$$\lim_{(x,y) \rightarrow (0,0)} \frac{xy}{(x^2 + y^2)^2} = \lim_{(x,x) \rightarrow (0,0)} \frac{x(x)}{(x^2 + (x)^2)^2} = \lim_{x \rightarrow 0} \frac{x^2}{4x^4} = \lim_{x \rightarrow 0} \frac{1}{4x^2} = \infty$$

Since this limit does not exist along the path $y = x$ we can say that $\lim_{(x,y) \rightarrow (0,0)} \frac{xy}{(x^2 + y^2)^2} DNE$.

e) $\lim_{(x,y) \rightarrow (0,0)} \frac{xy - x^2y}{x^2 + y^2}$

If we progress towards $(0,0)$ along $y = 0$ we see that

$$\lim_{(x,y) \rightarrow (0,0)} \frac{xy - x^2y}{x^2 + y^2} = \lim_{(x,0) \rightarrow (0,0)} \frac{x(0) - x^2(0)}{x^2 + (0)^2} = \lim_{x \rightarrow 0} \frac{0}{x^2} = \lim_{x \rightarrow 0} 0 = 0.$$

However, if we progress towards $(0,0)$ along $y = x$ we see that

$$\lim_{(x,y) \rightarrow (0,0)} \frac{xy - x^2y}{x^2 + y^2} = \lim_{(x,x) \rightarrow (0,0)} \lim_{(x,y) \rightarrow (0,0)} \frac{x(x) - x^2(x)}{x^2 + (x)^2} = \lim_{x \rightarrow 0} \frac{x^2 - x^3}{2x^2} = \lim_{x \rightarrow 0} \frac{1 - x}{2} = \frac{1}{2}$$

Since this limit has different answers as we progress towards $(0,0)$ along different paths, we can say

that $\lim_{(x,y) \rightarrow (0,0)} \frac{y^4 \cos^2(x)}{x^4 + y^4} DNE$.

f) $\lim_{(x,y) \rightarrow (1,1)} \frac{y^2}{x^2 + y^2}$

Since $f(x,y)$ is continuous at $(1,1)$ we easily compute that $\lim_{(x,y) \rightarrow (1,1)} \frac{y^2}{x^2 + y^2} = \frac{1^2}{1^2 + 1^2} = \frac{1}{2}$

g) $\lim_{(x,y) \rightarrow (1,1)} \frac{x^2 + y^2}{\sqrt{x^2 + y^2 + 1} - 1}.$

Note that

$$\begin{aligned} \lim_{(x,y) \rightarrow (1,1)} \frac{x^2 + y^2}{\sqrt{x^2 + y^2 + 1} - 1} &= \lim_{(x,y) \rightarrow (1,1)} \frac{x^2 + y^2}{\sqrt{x^2 + y^2 + 1} - 1} \cdot \frac{\sqrt{x^2 + y^2 + 1} + 1}{\sqrt{x^2 + y^2 + 1} + 1} = \lim_{(x,y) \rightarrow (1,1)} \frac{(x^2 + y^2)(\sqrt{x^2 + y^2 + 1} + 1)}{(x^2 + y^2 + 1) - 1} \\ &= \lim_{(x,y) \rightarrow (1,1)} \sqrt{x^2 + y^2 + 1} + 1 = \sqrt{1 + 1 + 1} + 1 \\ &= \sqrt{3} + 1 \end{aligned}$$

3. Consider the following piecewise functions $f(x, y)$ and determine $\lim_{(x,y) \rightarrow (0,0)} f(x, y).$

a) $f(x, y) = \begin{cases} \frac{xy}{x^2 + y^2} & \text{if } y \neq x^2 \\ \frac{x^3}{x^2 + 1} & \text{if } y = x^2 \end{cases}$

We see that along the path $y = x$,

$$\lim_{(x,y) \rightarrow (0,0)} f(x, y) = \lim_{(x,y) \rightarrow (0,0)} \frac{xy}{x^2 + y^2} = \lim_{(x,x) \rightarrow (0,0)} \frac{x(x)}{x^2 + (x)^2} = \lim_{(x,x) \rightarrow (0,0)} \frac{x^2}{2x^2} = \lim_{(x,x) \rightarrow (0,0)} \frac{1}{2} = \frac{1}{2}$$

We see that along the path $y = x^2$,

$$\lim_{(x,y) \rightarrow (0,0)} f(x, y) = \lim_{(x,y) \rightarrow (0,0)} \frac{x^3}{x^2 + 1} = \lim_{(x,x^2) \rightarrow (0,0)} \frac{x(x^2)}{x^2 + (x^2)^2} = \lim_{(x,x^2) \rightarrow (0,0)} \frac{x^3}{x^2 + x^4} = \lim_{(x,x^2) \rightarrow (0,0)} \frac{x}{1 + x^2} = 0$$

Clearly $\frac{1}{2} \neq 0$ and so $\lim_{(x,y) \rightarrow (0,0)} f(x, y)$ DNE.

b) $f(x, y) = \begin{cases} \frac{x^2 y^2}{x^2 y^2 + (x - y)^2} & \text{if } y \neq x \\ \frac{x^2}{x^2 + 1} & \text{if } y = x \end{cases}$ *Hint: Consider the path $y = x + x^2$ at some point.*

We see that along the path $y = x$, $\lim_{(x,y) \rightarrow (0,0)} f(x, y) = \lim_{(x,y) \rightarrow (0,0)} \frac{x^2}{x^2 + 1} = \lim_{(x,x) \rightarrow (0,0)} \frac{x^2}{x^2 + 1} = \frac{0}{0+1} = 0$

We see that along the path $y = x + x^2$,

$$\begin{aligned} \lim_{(x,y) \rightarrow (0,0)} f(x, y) &= \lim_{(x,y) \rightarrow (0,0)} \frac{x^2 y^2}{x^2 y^2 + (x - y)^2} = \lim_{(x, x+x^2) \rightarrow (0,0)} \frac{x^2 (x + x^2)^2}{x^2 (x + x^2)^2 + (x - (x + x^2))^2} \\ &= \lim_{(x, x+x^2) \rightarrow (0,0)} \frac{x^2 (x^2 + 2x^3 + x^4)}{x^2 (x^2 + 2x^3 + x^4) + x^4} = \lim_{(x, x+x^2) \rightarrow (0,0)} \frac{x^2 + 2x^3 + x^4}{(x^2 + 2x^3 + x^4) + x^2} \\ &= \lim_{(x, x+x^2) \rightarrow (0,0)} \frac{x^2 + 2x^3 + x^4}{2x^2 + 2x^3 + x^4} = \lim_{(x, x+x^2) \rightarrow (0,0)} \frac{1 + 2x + x^2}{2 + 2x + x^2} = \frac{1}{2} \end{aligned}$$

Clearly $\frac{1}{2} \neq 0$ and so $\lim_{(x,y) \rightarrow (0,0)} f(x, y)$ DNE.

c) $f(x, y) = \begin{cases} \frac{x^2 y^2 + 1}{x^3 + y^3 + 1} & \text{if } x^3 + y^3 \neq 0 \\ x^2 - xy & \text{if } x^3 + y^3 = 0 \end{cases}$

We see that along the path $y = 0$,

$$\lim_{(x,y) \rightarrow (0,0)} f(x, y) = \lim_{(x,y) \rightarrow (0,0)} \frac{x^2 y^2 + 1}{x^3 + y^3 + 1} = \lim_{(x,0) \rightarrow (0,0)} \frac{x^2 (0)^2 + 1}{x^3 + (0)^3 + 1} = \lim_{(x,0) \rightarrow (0,0)} \frac{1}{x^3 + 1} = \frac{1}{0+1} = 1$$

We see that along the path $y^3 = -x^3 \rightarrow y = -x$,

$$\lim_{(x,y) \rightarrow (0,0)} f(x, y) = \lim_{(x,y) \rightarrow (0,0)} x^2 - xy = \lim_{(x,-x) \rightarrow (0,0)} x^2 - x(-x) = \lim_{(x,-x) \rightarrow (0,0)} 2x^2 = 0$$

Clearly $1 \neq 0$ and so $\lim_{(x,y) \rightarrow (0,0)} f(x, y)$ DNE.

$$d) \quad f(x, y) = \begin{cases} \frac{x^3 + y^3}{x^2 + y^2} & \text{if } y \neq x^3 \\ \frac{2x^2y + 1}{x^4 + y^2} & \text{if } y = x^3 \end{cases}$$

We see that along the path $y = 0$,

$$\lim_{(x,y) \rightarrow (0,0)} f(x, y) = \lim_{(x,y) \rightarrow (0,0)} \frac{x^3 + y^3}{x^2 + y^2} = \lim_{(x,0) \rightarrow (0,0)} \frac{x^3 + (0)^3}{x^2 + (0)^2} = \lim_{(x,0) \rightarrow (0,0)} \frac{x^3}{x^2} = \lim_{(x,0) \rightarrow (0,0)} x = 0$$

We see that along the path $y = x^3$,

$$\lim_{(x,y) \rightarrow (0,0)} f(x, y) = \lim_{(x,y) \rightarrow (0,0)} \frac{2x^2y + 1}{x^4 + y^2} = \lim_{(x,x^3) \rightarrow (0,0)} \frac{2x^2(x^3) + 1}{x^4 + (x^3)^2} = \lim_{(x,x^3) \rightarrow (0,0)} \frac{2x^5 + 1}{x^4 + x^6} = \lim_{(x,x^3) \rightarrow (0,0)} \frac{2x + 1}{1 + x^2} = \frac{1}{1} = 1$$

Clearly $1 \neq 0$ and so $\lim_{(x,y) \rightarrow (0,0)} f(x, y)$ DNE.

$$e) \quad f(x, y) = \begin{cases} \frac{x^2 + y^2}{x^2 + xy + y^2} & \text{if } y \neq -x \\ \frac{x^2 - xy + y^2}{x^2 + y^2 + 1} & \text{if } y = x \end{cases}$$

We see that along the path $y = 0$,

$$\lim_{(x,y) \rightarrow (0,0)} f(x, y) = \lim_{(x,y) \rightarrow (0,0)} \frac{x^2 + y^2}{x^2 + xy + y^2} = \lim_{(x,0) \rightarrow (0,0)} \frac{x^2 + (0)^2}{x^2 + x(0) + (0)^2} = \lim_{(x,0) \rightarrow (0,0)} \frac{x^2}{x^2} = \lim_{(x,0) \rightarrow (0,0)} 1 = 1$$

We see that along the path $y = x$,

$$\lim_{(x,y) \rightarrow (0,0)} f(x, y) = \lim_{(x,y) \rightarrow (0,0)} \frac{x^2 - xy + y^2}{x^2 + y^2 + 1} = \lim_{(x,x) \rightarrow (0,0)} \frac{x^2 - x(x) + (x)^2}{x^2 + (x)^2 + 1} = \lim_{(x,x) \rightarrow (0,0)} \frac{x^2}{2x^2 + 1} = \frac{0}{0 + 1} = 0$$

Clearly $1 \neq 0$ and so $\lim_{(x,y) \rightarrow (0,0)} f(x, y)$ DNE.

4. Let F be a function of x and y . Consider the following statements.
- A. $f(x, y) \rightarrow 0$ as $(x, y) \rightarrow (0, 0)$ along every straight line through $(0, 0)$, but $\lim_{(x, y) \rightarrow (0, 0)} f(x, y)$ does not exist.
- B. $f(x, y) \rightarrow 0$ as $(x, y) \rightarrow (0, 0)$ along the lines $x = 0$ and $y = 0$, but $\lim_{(x, y) \rightarrow (0, 0)} f(x, y)$ does not exist.
- C. $f(x, y) \rightarrow 0$ as $(x, y) \rightarrow (0, 0)$ along the lines $x = 0$ and $y = 0$, but $\lim_{(x, y) \rightarrow (0, 0)} f(x, y) = 1$.

Which statements could *possibly* be true?

A could be true as long as $\lim_{(x, y) \rightarrow (0, 0)} f(x, y)$ does not exist along some other curve.

B could be true as long as $\lim_{(x, y) \rightarrow (0, 0)} f(x, y)$ does not exist along some other curve.

C *cannot* be true since along some paths $\lim_{(x, y) \rightarrow (0, 0)} f(x, y) = 0$ and so it cannot be true that

$$\lim_{(x, y) \rightarrow (0, 0)} f(x, y) = 1.$$

Therefore, only A & B could *possibly* be true.

5. Mark exactly one box corresponding to the correct ending of the sentence.

“The limit $\lim_{(x, y) \rightarrow (0, 0)} \frac{x^2 y^2}{x^4 + y^4} \dots$



... does not exist because the limits as one approaches $(0, 0)$ along the lines $x = 0$ and $y = x$ are different.”



... does not exist because the limits as one approaches $(0, 0)$ along the curves $y = x^2$ and $x = y^2$ are different.”



... exists because $\frac{x^2 y^2}{x^4 + y^4}$ is a composition of continuous functions”



... exists because the limits as one approaches $(0, 0)$ along the lines $y = x$ and $y = -x$ are the same.”

The 2nd statement is false since along both of the given paths $\lim_{(x, y) \rightarrow (0, 0)} \frac{x^2 y^2}{x^4 + y^4} = 0$.

The 3rd statement is false since the function is discontinuous at $(0, 0)$ and at least one function used in a composition to create this would also be discontinuous at $(0, 0)$.

The 4th statement is false since we have not determined that the limit is always produces the same result along *all* paths approaching $(0, 0)$.

6. Let f be a function from $\mathbb{R}^2$ to $\mathbb{R}$. Suppose that $f(x, y) \rightarrow 3$ as $(x, y) \rightarrow (0, 1)$ along every line of the form $y = kx + 1$. What can you say about the limit $\lim_{(x, y) \rightarrow (0, 1)} f(x, y)$? Select all that apply.

We cannot determine the limit at this time since we do not know what happens as $(x, y) \rightarrow (0, 1)$ along *all* possible paths.

☐

It exists and is equal to 3.

☒

We cannot determine if the limit exists, but if it does, the limit is 3.

☐

It does not exist.

7. Determine 5 different paths along which you can compute the limit $\lim_{(x, y) \rightarrow (2, 3)} f(x, y)$.

Since our limit has (x, y) approaching $(2, 3)$ we need paths that lead us to the point $(2, 3)$.

Here are 5 different possible paths:

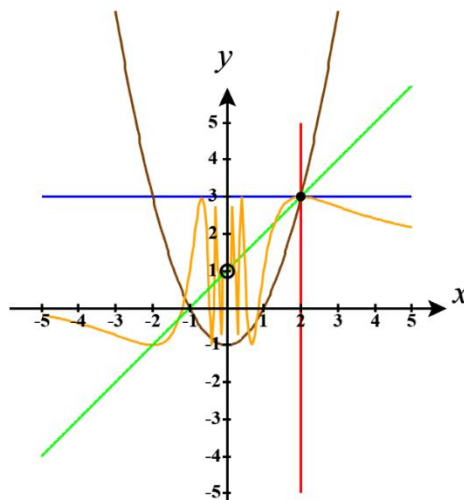
Path 1: $x = 2$

Path 2: $y = 3$

Path 3: $y = x + 1$

Path 4: $y = x^2 - 1$

Path 5: $2 \sin\left(\frac{\pi}{x}\right) + 1$



8. Calculate the following limit $\lim_{(x, y) \rightarrow (0, 0)} \frac{e^{-1/\sqrt{x^2 + y^2}}}{\sqrt{x^2 + y^2}}$.

If we let $t = x^2 + y^2$, then we see that as $(x, y) \rightarrow (0, 0)$ it is the case that $t \rightarrow 0^+$.

Therefore, we can say that $\lim_{(x, y) \rightarrow (0, 0)} \frac{e^{-1/\sqrt{x^2 + y^2}}}{\sqrt{x^2 + y^2}} = \lim_{t \rightarrow 0^+} \frac{e^{-1/\sqrt{t}}}{\sqrt{t}}$

Now since we have reduced the limit to a limit in one variable we see that using L'Hopital's Rule is valid. However, the current form of the limit will cause the use of L'Hopital's Rule to loop indefinitely and therefore we will first rewrite the limit and then calculate the result.

$$\lim_{t \rightarrow 0^+} \frac{e^{-1/\sqrt{t}}}{\sqrt{t}} = \lim_{t \rightarrow 0^+} \frac{t^{-1/2}}{e^{-1/\sqrt{t}}} = \lim_{t \rightarrow 0^+} \frac{t^{-1/2}}{e^{-t^{-1/2}}} \stackrel{H}{=} \lim_{t \rightarrow 0^+} \frac{-\frac{1}{2}t^{-3/2}}{\frac{1}{2}t^{-3/2}e^{-t^{-1/2}}} = \lim_{t \rightarrow 0^+} \frac{-1}{e^{-t^{-1/2}}} = 0$$

Therefore, $\lim_{(x, y) \rightarrow (0, 0)} \frac{e^{-1/\sqrt{x^2 + y^2}}}{\sqrt{x^2 + y^2}} = 0$.

9. Is the function $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ continuous at $(0,0)$?

$$\text{a) } f(x, y) = \begin{cases} x^2 + y + 1 & \text{if } (x, y) \neq (0, 0) \\ 0 & \text{if } (x, y) = (0, 0) \end{cases}$$

We see that $\lim_{(x,y) \rightarrow (0,0)} f(x, y) = \lim_{(x,y) \rightarrow (0,0)} (x^2 + y + 1) = 1$ and $f(0,0) = 0$. Therefore,

$\lim_{(x,y) \rightarrow (0,0)} f(x, y) \neq f(0,0)$ and thus f is not continuous at $(0,0)$.

$$\text{b) } f(x, y) = \begin{cases} \frac{x^2 y^3}{2x^2 + y^2} & \text{if } (x, y) \neq (0, 0) \\ 1 & \text{if } (x, y) = (0, 0) \end{cases}$$

Along $y=0$ we get $\lim_{(x,y) \rightarrow (0,0)} f(x, y) = \lim_{(x,0) \rightarrow (0,0)} \frac{x^2 (0)^3}{2x^2 + (0)^2} = \lim_{(x,0) \rightarrow (0,0)} 0 = 0$

Note that this implies that either $\lim_{(x,y) \rightarrow (0,0)} f(x, y) = 0$ or $\lim_{(x,y) \rightarrow (0,0)} f(x, y)$ DNE.

However, $f(0,0) = 1$ and so there is no way that $\lim_{(x,y) \rightarrow (0,0)} f(x, y) = f(0,0)$. Therefore, $f(x, y)$ cannot be continuous at $(0,0)$, but will be continuous everywhere else.

$$\text{c) } f(x, y) = \begin{cases} \frac{xy}{x^2 + xy + y^2} & \text{if } (x, y) \neq (0, 0) \\ 0 & \text{if } (x, y) = (0, 0) \end{cases}$$

Along $y=0$ we get $\lim_{(x,y) \rightarrow (0,0)} f(x, y) = \lim_{(x,0) \rightarrow (0,0)} \frac{x(0)}{x^2 + xy + y^2} = \lim_{(x,0) \rightarrow (0,0)} 0 = 0$.

Along $y=x$ we get $\lim_{(x,y) \rightarrow (0,0)} f(x, y) = \lim_{(x,0) \rightarrow (0,0)} \frac{x^2}{x^2 + x^2 + x^2} = \lim_{(x,0) \rightarrow (0,0)} \frac{x^2}{3x^2} = \frac{1}{3}$

Note that this implies that either $\lim_{(x,y) \rightarrow (0,0)} f(x, y)$ DNE.

Therefore, $f(x, y)$ cannot be continuous at $(0,0)$, but will be continuous everywhere else.